

Units and Dimensions

Single Correct Type Questions

1. Match List-I with List-II.

List-I		List-II	
A.	Young's Modulus (Y)	I.	$[ML^{-1} T^{-1}]$
B.	Coefficient of Viscosity (η)	II.	$[ML^2 T^{-1}]$
C.	Planck's Constant (h)	III.	$[ML^{-1} T^{-2}]$
D.	Work Function (ϕ)	IV.	$[ML^2 T^{-2}]$

Choose the correct answer from the options given below:

[25 Jan, 2023 (Shift-II)]

- (a) A-II, B-III, C-IV, D-I
- (b) A-III, B-I, C-II, D-IV
- (c) A-I, B-III, C-IV, D-II
- (d) A-I, B-II, C-III, D-IV

2. Match List-I with List-II.

List-I		List-II	
A.	Angular momentum	I.	$[ML^2 T^{-2}]$
B.	Torque	II.	$[ML^{-2} T^{-2}]$
C.	Stress	III.	$[ML^2 T^{-1}]$
D.	Pressure gradient	IV.	$[ML^{-1} T^{-2}]$

Choose the correct answer from the options given below:

[31 Jan, 2023 (Shift-II)]

- (a) A-I, B-IV, C-III, D-II
- (b) A-III, B-I, C-IV, D-II
- (c) A-II, B-III, C-IV, D-I
- (d) A-IV, B-II, C-I, D-III

3. Match List-I with List-II. [27 July, 2021 (Shift-II)]

List-I		List-II	
A.	Capacitance, C	I.	$M^1 L^1 T^{-3} A^{-1}$
B.	Permittivity of free space, ϵ_0	II.	$M^{-1} L^{-3} T^4 A^2$
C.	Permeability of free space, μ_0	III.	$M^{-1} L^{-2} T^4 A^2$
D.	Electric field, E	IV.	$M^1 L^1 T^{-2} A^{-2}$

Choose the correct answer from the options given below:

- (a) A-III, B-II, C-IV, D-I
- (b) A-III, B-IV, C-II, D-I
- (c) A-IV, B-II, C-III, D-I
- (d) A-IV, B-III, C-II, D-I

4. $\left(P + \frac{a}{V^2}\right)(V - b) = RT$ represents the equation of state of some gases. Where P is the pressure, V is the volume, T is the temperature and a , b , R are the constants. The physical

quantity, which has dimensional formula as that of $\frac{b^2}{a}$, will

be:

[1 Feb, 2023 (Shift-I)]

- (a) Bulk modulus
- (b) Modulus of rigidity
- (c) Compressibility
- (d) Energy density

5. The frequency (ν) of an oscillating liquid drop may depend upon radius (r) of the drop, density (ρ) of liquid and the surface tension (s) of the liquid as: $\nu = r^a \rho^b s^c$. The values of a , b and c respectively are [24 Jan, 2023 (Shift-II)]

- (a) $\left(-\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}\right)$
- (b) $\left(\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}\right)$
- (c) $\left(\frac{3}{2}, \frac{1}{2}, -\frac{1}{2}\right)$
- (d) $\left(-\frac{3}{2}, \frac{1}{2}, \frac{1}{2}\right)$

6. In Vander Waals equation $\left[P + \frac{a}{V^2}\right][V - b] = RT$; P is pressure. V is volume, R is universal gas constant and T is temperature. The ratio of constants a/b is dimensionally equal to

[29 June, 2022 (Shift-I)]

- (a) P/V (b) V/P
(c) PV (d) PV^3

7. The work done by a gas molecule in an isolated system is given by $W = \alpha\beta^2 e^{-\frac{x^2}{\alpha kT}}$, where x is the displacement, k is the Boltzmann constant and T is the temperature. α and β are constants. Then the dimensions of β will be:

[24 Feb, 2021 (Shift-I)]

- (a) $[MLT^{-2}]$ (b) $[ML^2T^{-2}]$
(c) $[M^2LT^2]$ (d) $[M^0LT^0]$

8. A quantity f is given by $f = \sqrt{\frac{hc^5}{G}}$ where c is speed of light, G universal gravitational constant and h is the Planck's constant. Dimension of f is that of:

[9 Jan, 2020 (Shift-I)]

- (a) energy (b) momentum
(c) area (d) volume

9. If momentum (P), area (A) and time (T) are taken to be the fundamental quantities then the dimensional formula for energy is

[2 Sep, 2020 (Shift-II)]

- (a) $\left[P^{\frac{1}{2}}AT^{-1}\right]$ (b) $[P^2AT^{-2}]$
(c) $\left[PA^{\frac{1}{2}}T^{-1}\right]$ (d) $[PA^{-1}T^{-2}]$

10. The force of interaction between two atoms is given by

$$F = \alpha\beta \exp\left(-\frac{x^2}{\alpha kt}\right); \text{ where } x \text{ is the distance, } k \text{ is the}$$

Boltzmann constant and T is temperature and α and β are two constants. The dimension of β is:

[11 Jan, 2019 (Shift-I)]

- (a) $[M^0L^2T^{-4}]$ (b) $[M^2LT^{-4}]$
(c) $[MLT^{-2}]$ (d) $[M^2L^2T^{-2}]$

11. Two resistances are given as $R_1 = (10 \pm 0.5)\Omega$ and $R_2 = (15 \pm 0.5)\Omega$. The percentage error in the measurement of equivalent resistance when they are connected in parallel is

[6 Apr, 2023 (Shift-I)]

- (a) 6.33 (b) 2.33
(c) 4.33 (d) 5.33

12. If $Z = \frac{A^2B^3}{C^4}$, then the relative error in Z will be:

[25 June, 2022 (Shift-I)]

- (a) $\frac{\Delta A}{A} + \frac{\Delta B}{B} + \frac{\Delta C}{C}$
(b) $\frac{2\Delta A}{A} + \frac{3\Delta B}{B} - \frac{4\Delta C}{C}$
(c) $\frac{2\Delta A}{A} + \frac{3\Delta B}{B} + \frac{4\Delta C}{C}$
(b) $\frac{\Delta A}{A} + \frac{\Delta B}{B} - \frac{\Delta C}{C}$

13. A silver wire has a mass $(0.6 \pm 0.006)\text{g}$, radius $(0.5 \pm 0.005)\text{mm}$ and length $(4 \pm 0.04)\text{cm}$. The maximum percentage error in the measurement of its density will be:

[27 June, 2022 (Shift-I)]

- (a) 4% (b) 3%
(c) 6% (d) 7%

14. A student measuring the diameter of a pencil of circular cross-section with the help of a vernier scale records the following four readings 5.50 mm, 5.55 mm, 5.45 mm, and 5.65 mm. The average of these four readings is 5.5375 mm and the standard deviation of the data is 0.07395 mm. The average diameter of the pencil should therefore be recorded as

[6 Sep, 2022 (Shift-II)]

- (a) $(5.5375 \pm 0.0739)\text{mm}$
(b) $(5.54 \pm 0.07)\text{mm}$
(c) $(5.538 \pm 0.074)\text{mm}$
(d) $(5.5375 \pm 0.0740)\text{mm}$

15. In the experiment of Ohm's law, a potential difference of 5.0V is applied across the end of a conductor of length 10.0 cm and diameter of 5.00 mm. The measured current in the conductor is 2.00 A. The maximum permissible percentage error in the resistivity of the conductor is:

[18 March, 2021 (Shift-I)]

- (a) 3.9 (b) 8.4
(c) 3.0 (d) 7.5

16. The time period of a simple pendulum is given by

$$T = 2\pi\sqrt{\frac{l}{g}}. \text{ The measured value of the length of pendulum}$$

is 10 cm known to a 1 mm accuracy. The time for 200 oscillations of the pendulum is found to be 100 second using a clock of 1 s resolution. The percentage accuracy in the determination of ' g ' using this pendulum is ' x '. The value of ' x ' to be nearest integer is

[18 March, 2021 (Shift-I)]

- (a) 2% (b) 3%
(c) 5% (d) 4%

17. A physical 'y' is represented by the formula $y = m^2 r^{-4} g^x \ell^{-\frac{3}{2}}$ if the percentage errors found in y, m, r, ℓ and g are 18, 1, 0.5, 4 and p respectively, then find the value of x and p .
[27 July, 2021 (Shift-II)]

- (a) 5 and ± 2 (b) $\frac{16}{3}$ and $\pm \frac{3}{2}$
(c) 8 and ± 2 (d) 4 and ± 3

18. In a simple pendulum experiment for determination of acceleration due to gravity (g), time taken for 20 oscillations is measured by using a watch of 1 second least count. The mean value of time taken comes out to be 30s. The length of pendulum is measured by using a meter scale of least count 1 mm and the value obtained is 55.0 cm. The percentage error in the determination of g is close to:
[8 April, 2019 (Shift-II)]

- (a) 0.7% (b) 3.5%
(c) 6.8% (d) 0.2%

19. The diameter and height of a cylinder are measured by a meter scale to be 12.6 ± 0.1 cm and 34.2 ± 0.1 cm respectively. What will be the value of its volume in appropriate significant figures? [10 Jan, 2019 (Shift-II)]

- (a) 4264 ± 81 cm³ (b) 4264 ± 81.0 cm³
(c) 4260 ± 80 cm³ (d) 4300 ± 80 cm³

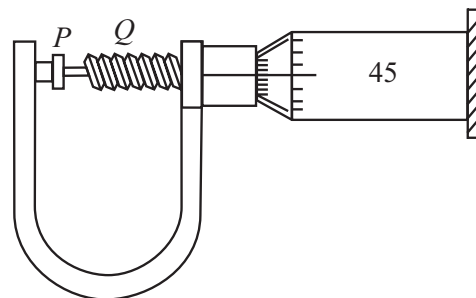
20. In an experiment with Vernier callipers of least count 0.1 mm, when two jaws are joined together the zero of Vernier scale lies right to the zero of the main scale and 6th division of Vernier scale coincides with the main scale division. While measuring the diameter of a spherical bob, the zero of vernier scale lies in between 3.2 cm and 3.3 cm marks, and 4th division of vernier scale coincides with the main scale division. The diameter of bob is measured as :
[10 Apr, 2023 (Shift-II)]

- (a) 3.18 cm (b) 3.25 cm
(c) 3.26 cm (d) 3.22 cm

21. In a Vernier Calliper, 10 divisions of Vernier scale is equal to the 9 divisions of main scale. When both jaws of Vernier calipers touch each other, the zero of the Vernier scale is shifted to the left of zero of the main scale and 4th Vernier scale division exactly coincides with the main scale reading. One main scale division is equal to 1 mm. While measuring diameter of a spherical body, the body is held between two jaws. It is now observed that zero of the Vernier scale lies between 30 and 31 divisions of main scale reading and 6th Vernier scale division exactly coincides with the main scale reading. The diameter of the spherical body will be [26 July, 2022 (Shift-II)]

- (a) 3.02 cm (b) 3.06 cm
(c) 3.10 cm (d) 3.20 cm

22. In an experiment to find out the diameter of wire using screw gauge, the following observations were noted?



- A. Screw moves 0.5 mm or main scale in one complete rotation.
B. Total divisions on circular scale = 50
C. Main scale reading is 2.5 mm
D. 45th division of circular scale is in the pitch line
Then the diameter of wire is [29 July, 2022 (Shift-I)]

- (a) 2.92 mm
(b) 2.54 mm
(c) 2.98 mm
(d) 3.45 mm

23. A screw gauge of pitch 0.5 mm is used to measure the diameter of uniform wire of length 6.8cm, the main scale reading is 1.5 mm and circular scale reading is 7. The calculated curved surface area of wire to appropriate significant figures is:

[26 July, 2022 (Shift-I)]

[Screw gauge has 50 divisions on its circular scale]

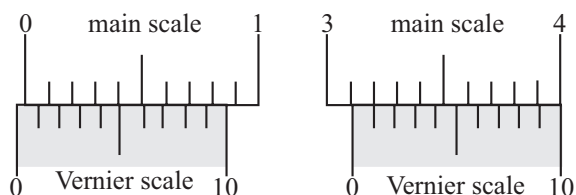
- (a) 6.8 cm²
(b) 3.4 cm²
(c) 3.9 cm²
(d) 2.4 cm²

24. The pitch of the screw gauge is 1 mm and there are 100 divisions on the circular scale. When nothing is put in between the jaws, the zero of the circular scale lies 8 divisions below the reference line. When a wire is placed between the jaws, the first linear scale division is clearly visible while 72nd division on circular scale coincides with the reference line, The radius of the wire is:

[25 Feb, 2021 (Shift-I)]

- (a) 0.82 mm
(b) 1.64 mm
(c) 0.90 mm
(d) 1.80 mm

25. The smallest division on the main scale of a Vernier callipers is 0.1 cm. Ten divisions of the Vernier scale correspond to nine divisions of the main scale. The figure below on the left shows the reading of this calliper with no gap between its two jaws. The figure on the right shows the reading with a solid sphere held between the jaws. The correct diameter of the sphere is [JEE Adv, 2021]



- (a) 3.07 cm (b) 3.11 cm
(c) 3.15 cm (d) 3.17 cm

Integer Type Questions

26. The Vernier constant of Vernier callipers is 0.1 mm and it has zero error of (-0.05) cm. While measuring diameter of a sphere, the main scale reading is 1.7 cm and coinciding vernier division is 5. The corrected diameter will be $\times 10^{-2}$ cm. [29 June, 2022 (Shift-II)]
27. In experiment of find acceleration due to gravity (g) using simple pendulum, time period of 0.5 s is measured from time of 100 oscillation with a watch of 1 s resolution. If measured value of length is 10 cm known to 1 mm accuracy, The accuracy in the determination of g is found to be $x\%$. The value of x is _____.

[28 July, 2022 (Shift-II)]

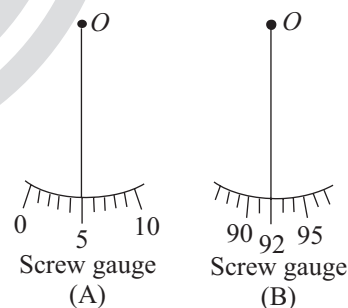
28. In a screw gauge, there are 100 divisions on the circular scale and the main scale moves by 0.5 mm on a complete rotation of the circular scale. The zero of circular scale lies 6 divisions below the line of graduation when two studs are brought in contact with each other. When a wire is placed between the studs, 4 linear scale divisions are clearly visible while 46th division the circular scale coincide with the reference line. The diameter of the wire is $\times 10^{-2}$ mm. [30 Jan, 2023 (Shift-I)]

29. The one division of main scale of Vernier callipers reads 1 mm and 10 divisions of Vernier scale is equal to the 9 divisions on main scale. When the two jaws of the instrument touch each other, the zero of the Vernier lies to the right of zero of the main scale and its fourth division coincides with a main scale division. When a spherical bob is tightly placed between the two jaws, the zero of the Vernier scale lies in between 4.1 cm and 4.2 cm and 6th Vernier division coincides with a main scale division. The diameter of the bob will be $\times 10^{-2}$ cm

[27 July, 2022 (Shift-I)]

30. Student A and Student B used two screw gauges of equal pitch and 100 equal circular divisions to measure the radius of a given wire. The actual value of the radius of the wire is 0.322 cm. The absolute value of the difference between the final circular scale readings observed by the students A and B is _____.

[25 July, 2021 (Shift-I)]



[Figure shows position of reference 'O' when jaws of screw gauge are closed]

Given pitch = 0.1 cm.

ANSWER KEY

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|-----------|---------|-----------|-----------|----------|
| 1. (b) | 2. (b) | 3. (a) | 4. (c) | 5. (a) | 6. (c) | 7. (a) | 8. (a) | 9. (c) | 10. (b) |
| 11. (c) | 12. (c) | 13. (a) | 14. (b) | 15. (a) | 16. (b) | 17. (b) | 18. (c) | 19. (c) | 20. (a) |
| 21. (c) | 22. (a) | 23. (b) | 24. (a) | 25. (c) | 26. [180] | 27. [5] | 28. [220] | 29. [412] | 30. [13] |

EXPLANATIONS

$$1. (b) Y = \frac{\text{Stress}}{\text{Strain}} = \frac{[MLT^{-2}]}{[L^2]} = [ML^{-1}T^{-2}]$$

$$\therefore \eta = \frac{F}{6\pi r v}$$

$$\Rightarrow [\eta] = \frac{[MLT^{-2}]}{[L][LT^{-1}]} = [ML^{-1}T^{-1}]$$

$$\therefore h = \frac{E}{v}$$

$$\Rightarrow h = \frac{[ML^2T^{-2}]}{[T^{-1}]} = [ML^2T^{-1}]$$

Work function has same dimension as that of energy,
so $[\phi] = [ML^2T^{-2}]$

$$2. (b) [\text{Stress}] = \frac{[F]}{[A]} = \frac{[MLT^{-2}]}{L^2} = [ML^{-1}T^{-2}]$$

[Pressure gradient]

$$= \frac{[P]}{[Z]} = \frac{[ML^{-1}T^{-2}]}{L^1} = [ML^{-2}T^{-2}]$$

$$[\text{Torque}] = [r][F] = [L][MLT^{-2}] = [ML^2T^{-2}]$$

$$[\text{Angular Momentum}] = [\tau][t] = [ML^2T^{-2}][T] = [ML^2T^{-1}]$$

$$3. (a) \text{Capacitance } [C] = \frac{Q}{V} = \frac{Q}{W/Q} = \frac{Q^2}{W} = \frac{A^2T^2}{M^1L^2T^{-2}}$$

$$= M^{-1}L^{-2}T^4A^2$$

$$\text{Permittivity } [\epsilon_0] = M^{-1}L^{-3}T^4A^2$$

$$C = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \Rightarrow \mu_0 = \frac{1}{\epsilon_0 c^2}$$

$$\Rightarrow \text{Permeability of freespace } [\mu_0] = [M^1L^1T^{-2}A^{-2}]$$

$$\text{Electric field, } E = \frac{F}{q} = \frac{MLT^{-2}}{AT}$$

$$\Rightarrow [E] = MLT^{-3}A^{-1}$$

$$4. (c) b \text{ has the dimensions of volume, } [b] = [V]$$

$$\therefore \left[\frac{a}{V^2} \right] = [P]$$

$$\Rightarrow \left[\frac{a}{b^2} \right] = [P] \therefore \left[\frac{b^2}{a} \right] = \frac{1}{[P]} = \frac{1}{[B]} = [K]$$

$$\therefore \left[\frac{b^2}{a} \right] = \frac{1}{[P]} = \frac{1}{[B]} = [K]$$

$$5. (a) [T^{-1}] = [L^1]^a [M^1L^{-3}]^b \left[\frac{MLT^{-2}}{L} \right]^c$$

$$\Rightarrow [T^{-1}] = [M^{b+c} \cdot L^{a-3b} \cdot T^{-2c}]$$

Equating the power,

$$c = \frac{1}{2}, b = -\frac{1}{2}, a - 3b = 0$$

$$\Rightarrow a = -\frac{3}{2}$$

$$6. (c) \text{From principle of homogeneity,}$$

$$[P] = \left[\frac{a}{V^2} \right] \text{ and } [b] = [V]$$

$$\Rightarrow \left[\frac{a}{b} \right] = [PV]$$

7. (a) Since, $\frac{x^2}{\alpha k T}$ should be dimensionless.

$$\therefore [\alpha] = \frac{L^2}{ML^2 T^{-2}} = M^{-1} T^2$$

Dimension of $\alpha\beta^2$ should be dimension of W .

$$\text{So, } [\alpha\beta^2] = ML^2 T^{-2}$$

$$\Rightarrow [\beta^2] = \frac{ML^2 T^{-2}}{M^{-1} T^2} = M^2 L^2 T^{-4} \Rightarrow [\beta] = MLT^{-2}$$

8. (a) Planck's constant, $(h) = [M^1 L^2 T^{-1}]$

$$\text{Speed of light, } (c) = [L^1 T^{-1}]$$

$$\text{Universal gravitational constant } (G) = [M^{-1} L^2 T^{-2}]$$

$$(G) = \frac{\sqrt{M^1 L^2 T^{-1} \times L^5 \times T^{-5}}}{\sqrt{M^{-1} L^3 T^{-2}}} = M^1 L^2 T^{-2}$$

9. (c) Energy = Force $\times$ Distance

$$\begin{aligned} \Rightarrow [\text{Energy}] &= \frac{P}{T} \times \sqrt{A} \\ &= \left[PA^{\frac{1}{2}} T^{-1} \right] \end{aligned}$$

10. (b) Power of exponential should be dimensionless.

$$\therefore [x^2] = [\alpha k t]$$

$$\Rightarrow [L^2] = [\alpha] [ML^2 T^{-2}]$$

$$\Rightarrow [\alpha] = [M^{-1} T^2]$$

$$\therefore [F] = [\alpha] [\beta]$$

$$[MLT^{-2}] = [M^{-1} T^2] [\beta]$$

$$\therefore [\beta] = [M^2 LT^{-4}]$$

11. (c) When R_1 and R_2 are connected in parallel

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$$

... (i)

$$\left[R = \frac{R_1 R_2}{R_1 + R_2} = \frac{10 \times 15}{10 + 15} = 6 \right]$$

Differentiating (i) both sides

$$\frac{\Delta R}{R^2} = \frac{\Delta R_1}{R_1^2} + \frac{\Delta R_2}{R_2^2}$$

$$\Rightarrow \frac{\Delta R}{R} = \left(\frac{\Delta R_1}{R_1^2} + \frac{\Delta R_2}{R_2^2} \right) R$$

$$= \left(\frac{0.5}{100} + \frac{0.5}{225} \right) 6 \times \frac{13}{300}$$

$$\frac{\Delta R}{R} \times 100 = \frac{13}{3} = 4.33\%$$

$$12. (c) z = \frac{A^2 B^3}{C^4}$$

The relative error in z can be given as

$$\frac{\Delta z}{z} = \frac{2\Delta A}{A} + \frac{3\Delta B}{B} + \frac{4\Delta C}{C}$$

13. (a) Given, $m = 0.6 \pm 0.006$, $r = 0.5 \pm 0.005$, $l = 4 \pm 0.04$

$$\text{Percentage error in } m, \frac{\Delta m}{m} \times 100 = \frac{0.006}{0.6} \times 100 = 1\%$$

$$\text{Percentage error in } r, \frac{\Delta r}{r} \times 100 = \frac{0.005}{0.5} \times 100 = 1\%$$

$$\text{Percentage error in } l, \frac{\Delta l}{l} \times 100 = \frac{0.04}{4} \times 100 = 1\%$$

Formula of density

$$\Rightarrow \rho = \frac{m}{V} = \frac{m}{\pi r^2 l}$$

$$\therefore \frac{\Delta \rho}{\rho} \times 100 = \frac{\Delta m}{m} \times 100 + 2 \frac{\Delta r}{r} \times 100 + \frac{\Delta l}{l} \times 100$$

$$= 1 + 2 \times 1 + 1 = 4\%.$$

$$14. (b) d_{av} = 5.5375 \text{ mm}$$

$$\Delta d = 0.07395 \text{ mm}$$

$\therefore$ Measured data are up to two digits after decimal

$$\therefore d = (5.54 \pm 0.07) \text{ mm}$$

$$15. (a) V = IR = I \frac{\rho \ell}{\pi \frac{d^2}{4}} \Rightarrow \rho = \frac{\pi d^2 V}{4 \ell I}$$

$$\text{So, } \frac{\Delta \rho}{\rho} = 2 \cdot \frac{\Delta d}{d} + \frac{\Delta V}{V} + \frac{\Delta \ell}{\ell} + \frac{\Delta I}{I}$$

$$= 2 \times \frac{0.01}{5.00} + \frac{0.1}{5.0} + \frac{0.1}{10.0} + \frac{0.01}{2000} = 0.039$$

$$\text{Percentage error} = \frac{\Delta \delta}{\delta} \times 100 = 3.9\%$$

$$16. (b) T = 2\pi \sqrt{\frac{l}{g}} \Rightarrow T^2 = 2\pi \left(\frac{l}{g} \right) \Rightarrow g = 2\pi \frac{l}{T^2}$$

$$\Rightarrow \frac{\Delta g}{g} = \frac{\Delta l}{l} + \frac{2\Delta T}{T}$$

$$\Rightarrow \frac{\Delta g}{g} = \frac{1 \times 10^{-3}}{1 \times 10^{-2}} + \frac{2 \times 1}{100}$$

$$\Rightarrow \frac{\Delta g}{g} = 0.02 + 0.01 = 0.03$$

$$\Rightarrow 100 \times \frac{\Delta g}{g} = 0.03 \times 100 = 3\%$$

$$17. (b) \left(\frac{\Delta y \times 100}{y} \right) = \left[2 \times \frac{\Delta m}{m} + \frac{4 \times \Delta r}{r} + \frac{x \times \Delta g}{g} + \frac{3}{2} \times \frac{\Delta \ell}{\ell} \right] \times 100$$

$$\Rightarrow 18 = (2 \times 1) + (4 \times 0.5) + (xp) + \left(\frac{3}{2} \times 4 \right)$$

$$\Rightarrow 8 = xp$$

Using given options

$$\Rightarrow x = \frac{16}{3}; p = \pm \frac{3}{2}$$

18. (c) In case of simple pendulum,

$$T = 2\pi \sqrt{\frac{L}{g}}$$

$$\therefore g = \frac{4\pi^2 L}{T^2}$$

$$\% \text{ error in } g = \frac{\Delta g}{g} \times 100 = \left(\frac{\Delta L}{L} + \frac{2\Delta T}{T} \right) 100\%$$

$$= \left(\frac{0.1}{55} + 2 \left(\frac{1}{20} \right) \right) 100\% \approx 6.8\%$$

$$19. (c) V = \pi R^2 h = \frac{\pi}{4} D^2 h = 4260 \text{ cm}^3$$

$$\therefore \frac{\Delta V}{V} = 2 \frac{\Delta D}{D} + \frac{\Delta h}{h}$$

$$\Delta V = \left(2 \times \frac{0.1}{12.6} + \frac{0.1}{34.2} \right) V$$

$$= \frac{2 \times 426}{12.6} + \frac{426}{34.2}$$

$$= 67.61 + 12.459 = 80.075$$

$$\therefore V = 4260 \pm 80 \text{ cm}^3$$

20. (a) $\therefore$ Zero Error = $6 \times$ Least Count = 0.6 mm

$$\therefore \text{Reading} = \text{MSR} + \text{VSR} \times \text{LC} - \text{Zero Error}$$

$$= [32 \text{ mm} + (0.1)4\text{mm}] - 0.6 \text{ mm}$$

$$= 31.8 \text{ mm} = 31.8 \times 10^{-1} \text{ cm} = 3.18 \text{ cm}$$

$$= 3.18 \text{ cm}$$

21. (c) 1 M.S.D. = 1 mm

Now, 9 M.S.D. = 10 V.S.D.

$$1 \text{ V.S.D.} = 0.9 \text{ M.S.D.} = 0.9 \text{ mm} = 0.01 \text{ cm}$$

$$\text{Now, L.C. of vernier caliper} = 1 - 0.9 = 0.1 \text{ mm}$$

$$= 0.01 \text{ cm}$$

$$\text{zero error} = -(10 - 4) \times 0.1 \text{ mm} = -0.6 \text{ mm}$$

$$\text{Reading} = \text{M.S.R} + \text{V.S.R} - \text{Zero error}$$

$$= 3 \text{ cm} + 6 \times 0.01 - [-0.06]$$

$$= 3 + 0.06 + 0.06$$

$$= 3.12 \text{ cm} \sim \text{nearest to } 3.10$$

$$22. (a) \text{ L.C.} = \frac{0.5}{50} \text{ mm} = 0.01 \text{ mm}$$

$$d = (2.5 + 45 \times 0.01) \text{ mm} = 2.95 \text{ mm}$$

23. (b) Least count of screw gauge

$$= \frac{\text{Pitch}}{\text{No. of division on circular scale}}$$

$$= \frac{0.5}{50} \text{ mm} = 0.01 \text{ mm}$$

$$\text{Diameter, } d = 1.5 + 7 \times 0.01$$

$$= 1.57 \text{ mm} = 0.157 \text{ cm}$$

$$\therefore \text{Curved Surface Area} = (2\pi r)\ell = \pi d\ell \quad (\because d = 2r)$$

$$= 3.142 \times 0.157 \times 6.8 = 3.354 \text{ cm}^2 = 3.4 \text{ cm}^2$$

$$24. (a) \text{ LC} = \frac{\text{Pitch}}{\text{No. of division}} = \frac{1}{100} \text{ mm} = 0.01 \text{ mm}$$

$$\text{Zero error} = +0.08 \text{ mm.}$$

$$\text{Diameter} = 1 + 72 \times 0.01 - 0.08 = 1.64 \text{ mm}$$

$$\text{Radius} = 0.82 \text{ mm}$$

25. (c) Given 10 VSD = 9 MSD

$$\text{Least count} = 1 \text{ VSD} = \frac{9}{10} \text{ MSD}$$

$$\text{Least count} = 1 \text{ MSD} - 1 \text{ VSD} = \left(1 - \frac{9}{10} \right) \text{ MSD}$$

$$= 0.1 \text{ MSD} = 0.1 \times 0.1 \text{ cm} = 0.01 \text{ cm}$$

As '0' of V.S. lie before '0' of M.S.

$$\text{Zero error} = -[10 - 6] \text{ L.C.}$$

$$= -4 \times 0.01 \text{ cm}$$

$$= -0.04 \text{ cm}$$

$$\text{Reading} = 3.1 \text{ cm} + 1 \times \text{LC} = 3.4 \text{ cm} + 1 \times 0.01 \text{ cm}$$

$$= 3.11 \text{ cm}$$

$$\begin{aligned}\text{True diameter} &= \text{Reading} - \text{Zero error} \\ &= 3.11 - (-0.04)\text{cm} = 3.15\text{ cm}\end{aligned}$$

26. [180] We have given, main scale reading = 1.7 cm
Zero correction = 0.05 cm
Vernier scale reading = $5 \times 0.1 = 0.5\text{ mm}$
The correction diameter of sphere = Main scale reading + Vernier scale reading + Zero correction
 $= 1.7 + 0.05 + 0.05 = 1.8\text{ cm} = 180 \times 10^{-2}\text{ cm}.$

27. [5] $T = 0.5\text{ sec}$

$$\text{No. of oscillation} = 100$$

$$\text{Resolution} = 1\text{sec}$$

$$\ell = 10\text{cm} \pm 1\text{cm}$$

$$T = 2\pi\sqrt{\frac{\ell}{g}} \Rightarrow g = \frac{4\pi^2\ell}{T^2}$$

$$\frac{dg}{g} \times 100 = \frac{d\ell}{\ell} \times 100 + \frac{2dT}{T} \times 100$$

$$= \frac{0.1}{10} \times 100 + \frac{2}{50} \times 100 = 5\%$$

$$\text{So, } x = 5$$

28. [220] Least count = $\frac{\text{Pitch}}{\text{No. of circular divisions}}$

$$= \frac{0.5\text{ mm}}{100}$$

$$\text{Least count} = 5 \times 10^{-3}\text{ mm}$$

$$\text{Positive Error} = \text{MSR} + \text{CSR}(\text{LC})$$

$$= 0\text{ mm} + 6(5 \times 10^{-3}\text{ mm})$$

$$\text{Reading of Diameter} = \text{MSR} + \text{CSR}(\text{LC}) -$$

$$\text{Positive zero error}$$

$$= 4 \times 0.5\text{ mm} + (46(5 \times 10^{-3})) - 6(5 \times 10^{-3})\text{ mm}$$

$$= 2\text{ mm} + 40 \times 5 \times 10^{-3}\text{ mm} = 2.2\text{ mm (Ans.)}$$

29.

$$[412] \text{ 1MSD} = 1\text{mm}$$

$$10\text{VSD} = 9\text{msD}$$

$$\text{Therefore, } 1\text{VSD} = 0.9\text{msD}$$

$$\text{L.C} = 1\text{MSD} - 1\text{VSD} = 1 - 0.9 = 0.1\text{mm}$$

$$\text{Positive zero error} = 4\text{LC} = 0.4\text{ mm}$$

$$\text{Reading} = \text{MSR} + \text{VSR} + \text{correction}$$

$$= 4.1\text{ cm} + 6 \times .01\text{cm} + (-0.04\text{cm})$$

$$= 4.12\text{cm} = 412 \times 10^{-2}\text{cm}$$

30.

$$[13] \text{ Difference in Reading}$$

$$= \text{Positive Zero Error} - \text{Negative Zero Error}$$

$$= (+5) - [-(100 - 92)] = 13$$